

Qatar-9b

R-2534

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-9_241225 · created 2026-08-25T23:25:35+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460669.8861 +/- 0.0074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.048
Transit depth (Rp/Rs)^2	1.99 +/- 1.35 [%]
Orbital Inclination (inc)	87.11 +/- 1.59
Airmass coefficient 1 (a1)	1.106 +/- 0.034
Airmass coefficient 2 (a2)	-0.069 +/- 0.056
Scatter in the residuals of the lightcurve fit is	3.09 %
Best Comparison Star	None
Optimal Aperture	2.94
Optimal Annulus	6.93
Transit Duration (day)	0.0723 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.141 ± 0.048 vs 0.1489 (deviation 0.15σ)
Duration fit vs published	0.0723 d vs 0.0779 d (deviation 0.79σ)
Mid-time O–C	0.7 min
Depth SNR	2.7
Residual scatter	3.09 %

Light curve

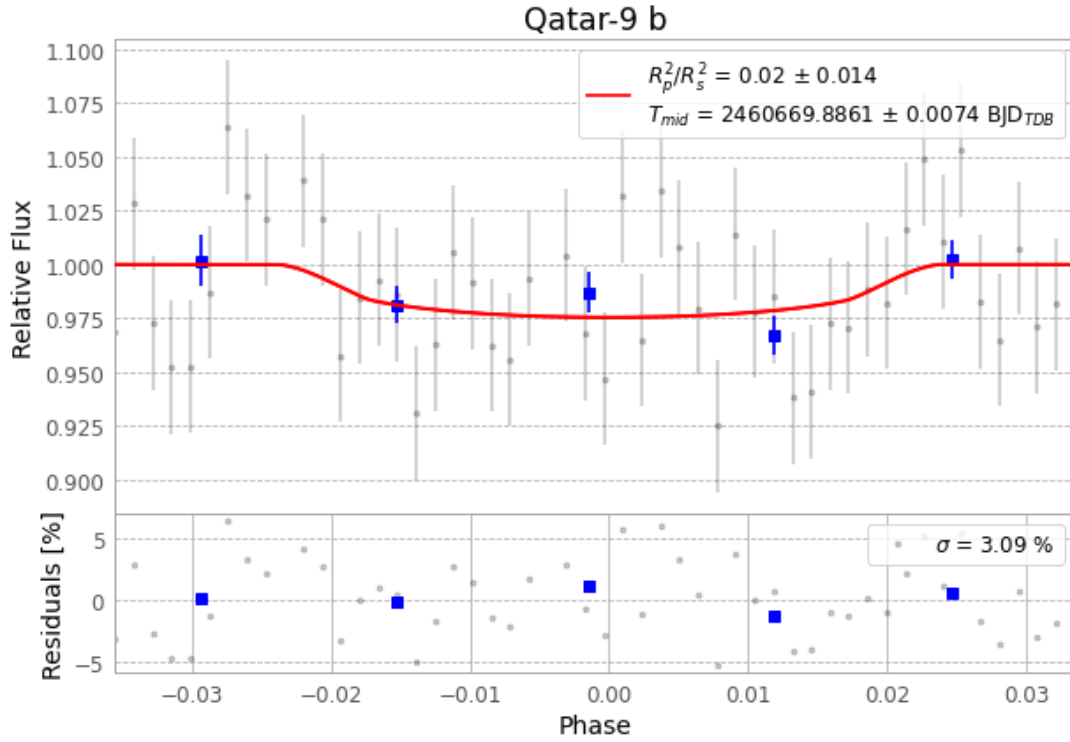


Figure 1 — FinalLightCurve_Qatar-9 b_2024-12-25.png (SHA-256 9ae7bbbeff76778c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 211], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 36e42f0f0e4555a5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

52 FITS frames (34 MB), Qatar-9241225075115.FITS → Qatar-9241225102415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-9-241225.log (16ca8f0fd933...), FinalLightCurve_Qatar-9 b_2024-12-25.png (9ae7bbbeff76...), FinalLightCurve_Qatar-9 b_2024-12-25.csv (b04f7bba0a03...)

Compute resources

Wall time 82.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-25 23:25Z.